Exercise Sheet 2

Due: 03.11.26 9h00

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General Regulations.

- Submission is optional but recommended; tutors provide feedback. Exercises are not graded.
- Work in groups of two or three, preferably from the same tutorial group. Submit all solutions as a single combined PDF via the Übungsgruppenverwaltung, once per group, and include the names of all group members.
- Solutions to tasks marked "solve yourself, without an AI agent" may be handwritten and scanned or typeset using LaTeX. Please ensure that scans are legible and not blurry.
- Course material is available in the GitHub repository.

1 Recap of probability theory

Solve this exercise yourself, without an AI agent.

You toss a fair coin twice, independently, and record the result of each toss in order. You are interested in the total number of heads.

(a) Outcomes, random variable and distribution. Write down the sample space Ω, i.e. all possible outcomes of the experiment, and assign a probability to each outcome. Define a random variable $X : \Omega \to \mathbb{R}$ describing the quantity of interest. For each outcome ω, state the value $X(\omega)$. List the possible values of X and determine their probabilities $p_X(x) = P(X = x)$.

(b) Expectation values. For a discrete random variable,

$$\mathbb{E}[g(X)] = \sum_x g(x)\, p_X(x),$$

where the sum runs over its possible values. Calculate $\mu = \mathbb{E}[X]$ and $\mathbb{E}[X^2]$ for your coin-toss experiment. Is $\mathbb{E}[X]$ a random variable or a fixed number?

(c) Variance identity. For any real-valued random variable Z with $\mathbb{E}[Z^2] < \infty$, prove that

$$\mathrm{Var}(Z) = \mathbb{E}\big[(Z - \mathbb{E}[Z])^2\big] = \mathbb{E}[Z^2] - \big(\mathbb{E}[Z]\big)^2.$$

You may use linearity of expectation: $\mathbb{E}[aU + bV] = a\mathbb{E}[U] + b\mathbb{E}[V]$ for constants a, b, and $\mathbb{E}[c] = c$ for any constant c. Use this result to evaluate the variance of the coin-toss variable X.

(d) Conditional probabilities. In the same experiment, define $Y = 1$ if the first toss is heads and $Y = 0$ otherwise. For each possible value x of X, identify the outcomes compatible with $X = x$ and determine

$$P(Y = 1 \mid X = x) \qquad \text{and} \qquad \mathbb{E}[Y \mid X = x].$$

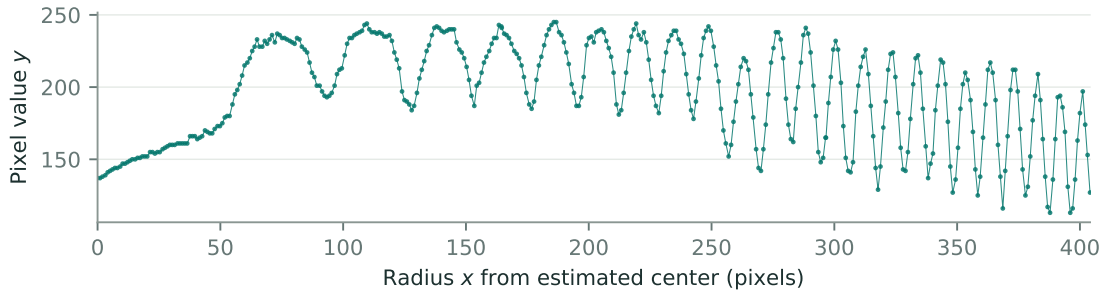
You may use $P(A \mid B) = P(A \cap B)/P(B)$ for $P(B) > 0$. For the conditional expectation, average the possible values of Y using these conditional probabilities.

2 Nadaraya–Watson regression

Use the agent only to write the required code in part (b). Solve parts (a) and (c) yourself, without an AI agent.

Measurements often follow complex, nonlinear trends whose shape is not known in advance. Nadaraya–Watson regression estimates such a trend by averaging nearby responses, giving closer observations more weight. Nadaraya–Watson regression is closely connected to kernel density estimation (KDE): it can be derived by using KDE to estimate the distribution of responses at a given input, then taking its mean. The inputs x_i in this exercise are fixed detector positions; the responses are measured pixel values.

The observations below show a radial profile of *Newton's rings*, an optical interference pattern. Locating bright and dark fringes is useful for measuring lens curvature. Here we explore smoothing the observations to estimate the unknown underlying intensity profile f.



Measured green-channel values at $N = 401$ fixed radial positions.
Data adapted from a photo by Winnie Summer, CC BY-SA 3.0.

Given observations (x_i, y_i), the prediction at x is

$$\widehat{m}_h(x) = \sum_{i=1}^{N} w_i(x) y_i, \qquad w_i(x) = \frac{K_h(x - x_i)}{\sum_{j=1}^{N} K_h(x - x_j)}.$$

Use the *biweight kernel*

$$K_h(t) = \begin{cases} \dfrac{15}{16h}\left(1 - \dfrac{t^2}{h^2}\right)^2, & |t| < h, \\ 0, & |t| \geq h, \end{cases} \qquad h > 0.$$

The bandwidth h is the radius of the neighborhood used for averaging.

The file `newton-rings.csv` in `sheet02/data` contains the measured data of the intensity profile. To compare sampling densities, use every fourth, every second, or every observation for $N = 101,\ 201,\ 401$, respectively; retain both endpoints.

(a) Reason about the estimator. Before writing any code, reason about the following questions:

(i) What happens if only one observation has non-zero weight? What if none does? What happens when h becomes much larger than the spacing between points?

(ii) Which bandwidths reproduce every measured value exactly? Would such a bandwidth also predict an observation well that was left out of the fit? Why or why not?

(iii) How does using every fourth, every second or every observation affect the range of useful bandwidths? Can a suitable bandwidth still resolve the most closely spaced fringes when only every fourth observation is used?

(iv) Can one bandwidth reproduce the slowly and rapidly oscillating parts equally well? Can predictions leave the range of observed responses?

(b) Implement and visualize. Use an AI agent to write code that produces plots and numerical results with which you can check your predictions from part (a) using the supplied data. Include the relevant results in your PDF.

(c) Interpret the results. Use the results from part (b) to assess your predictions from part (a). Explain your observations and discuss any discrepancies.

3 Linear regression

Use the agent only to write the required code in parts (a), (d) and (g). Solve parts (b), (c), (e), (f) and (h) yourself, without an AI agent.

The files `regression-plane.csv` and `regression-well.csv` in `sheet02/data` each contain $N = 300$ observations with two explanatory variables $\mathbf{x}_i = (x_{i1}, x_{i2})^\mathsf{T} \in \mathbb{R}^2$ and a scalar response $y_i \in \mathbb{R}$. The CSV columns are `x1`, `x2`, `y`. The responses contain measurement noise.

(a) Visualize the data. Use an AI agent to write code that loads both datasets and visualizes the observations as points in three dimensions, with axes x_1, x_2 and y.

(b) Inspect the data. Describe the shapes in part (a). For which dataset do you expect a plane to give a good fit?

(c) Derive the plane fit. Consider the model

$$f_{\boldsymbol{\beta}}(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2.$$

Write down the design matrix $X \in \mathbb{R}^{N\times 3}$, including the intercept column. Starting from $\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2$, derive the normal equations. When is the minimizer unique?

(d) Fit the planes. Use an AI agent to write code that fits the model from part (c) to each dataset. Report the fitted coefficients and the training mean squared error

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\big(y_i - f_{\widehat{\boldsymbol{\beta}}}(\mathbf{x}_i)\big)^2.$$

Plot the fitted planes together with the observations. Plot the residuals $r_i = y_i - f_{\widehat{\boldsymbol{\beta}}}(\mathbf{x}_i)$ against $x_{i1}^2 + x_{i2}^2$.

(e) Assess the fits. Which fit in part (d) leaves a systematic residual pattern, and what does it tell you about the model?

(f) Set up the squared-feature model. For `regression-well.csv`, use the model

$$g_{\boldsymbol{\theta}}(x_1, x_2) = \theta_0 + \theta_1 x_1^2 + \theta_2 x_2^2.$$

Write down the new design matrix.

(g) Fit the well using squared features. Use an AI agent to write code that fits the model from part (f) with the same least-squares method. Report its coefficients and MSE, and visualize its surface and residuals as in part (d).

(h) Compare and explain. Compare the fit in part (g) to the plane fitted to this dataset, using the surfaces, residuals and MSEs. Why is the fit in part (g) still called linear regression although its surface is curved? In which quantities is each model linear?